Supplementary file 4. Effects and outcomes of interventions on response to VAW during routine SRH consultation

Outcomes	Results (95% CI, p)	Effect (95% CI)	N of participants (design/follow up)	Study	Risk of bias	Direction of effect/interpretation
Direct effect on health-related cognition and emotions						
HCP knowledge about VAW, relevant procedures (n=4)						
Mean (SD) score of knowledge about domestic violence, post-intervention	Group training = 16.1 (1.9) CD training = 17.7 (1.1) p<0.001	-	35 HCPs (RCT/2 month)	Vakily 2017[58]	High	2-hour CD training improved HCP knowledge about domestic violence more than group training
Change in proportion with correct knowledge on sexual violence	Intervention = 31.4% Control = -22.3%	Net effect = 53.7% (32.2; 75.1)	HCPs (CBA/12 months	Abeid 2016[28]	High	5-day training, guidelines, infrastructure improvement improved HCPs knowledge about sexual violence
Median (IQR) score of knowledge about IPV	Pre-intervention = 0.62 (0.43-0.81) Post-intervention = 0.88 (0.82-0.94) p<0.001	-	408 HCPs (UBA/6 months)	Jayatilleke 2015[39]	High	4-day training, handbook, external referral improved HCP knowledge about IPV
Mean (95% CI) score of knowledge in providing care to sexual assault survivors	Pre-intervention = 49.09 (45.57; 51.34) Post-intervention = 61.59 (59.04; 64.42)	MD = 12.50 (10.29; 16.24)	106 HCPs (UBA/3 months)	Smith 2013[54]	High	4-day training, infrastructure improvement contributed towards improved HCP knowledge about providing clinical care to survivors of sexual violence
HCP attitudes about VAW (n=3)						
Mean (SD) score of attitudes about domestic violence, post-intervention	Group training = 46.9 (4.9) CD training = 45.4 (6.4) p = 0.3	-	35 HCPs (RCT/2 month)	Vakily 2017[58]	High	Neither group nor CD 2-hour training had effect on HCP attitudes about domestic violence
Mean (95% CI) score of attitudes about sexual violence	Pre-intervention = 71.76 (66.79; 73.14) Post-intervention = 77.20 (72.53; 78.34)	MD = 5.44 (1.89; 8.98)	106 HCPs (UBA/3 months)	Smith 2013[54]	High	4-day training, infrastructure improvement contributed towards improved attitudes about sexual violence
Change in proportion with accepting attitude towards sexual violence	Intervention = -4.1% Control = 6.8%	Net effect = -10.9% (-27.2; 5.5)	HCPs (CBA/12 months	Abeid 2016[28]	High	5-day training, guidelines, infrastructure improvement had no effect on HCP attitudes about sexual violence
HCP readiness for identifying and responding to VAW (n=2)						
Median (IQR) score of perceived barriers to IPV identification and response	Pre-intervention = 2.43 (2.14-3.14) Post-intervention = 1.14 (1.14-1.28) p<0.001		408 HCPs (UBA/6 months)	Jayatilleke 2015[39]	High	4-day training, handbook, external referral reduced HCP perceived barriers to identifying and responding to IPV
Median (IQR) score of perceived responsibilities to identify and respond to IPV	Pre-intervention = 3.20 (2.80-3.95) Post-intervention = 4.60 (4.20-4.80) p<0.001					4-day training, handbook increased HCP perceived responsibility and self-confidence to identify and respond to IPV

Outcomes	Results (95% CI, p)	Effect (95% CI)	N of participants (design/follow up)	Study	Risk of bias	Direction of effect/interpretation
Median (IQR) score of self-confidence to identify and respond to IPV	Pre-intervention = 1.81 (1.38-2.12) Post-intervention = 2.75 (2.62-2.88) p<0.001					
Mean (95% CI) score of HCPs' confidence in providing care to sexual assault survivors	Pre-intervention = 58.16 (53.86; 63.90) Post-intervention = 72.66 (66.21; 74.30)	MD = 14.50 (8.22; 20.77)	106 HCPs (UBA/3 months)	Smith 2013[54]	High	4-day training, infrastructure improvement contributed towards improved confidence in providing clinical care to sexual assault survivors
Women knowledge about VAW (n=1)						
Mean difference (95% CI) in women's IPV knowledge score, post-intervention	-	MD = 0.16 Crude β=0.176 (0.02; 0.033) Adjusted β=0.155 (0.00-0.31)	337 women (RCT/1 month)	Haberland 2016[38]	High	HCP training and ongoing support, 29-minute integrated HIV-IPV consultation, referral to IPV counsellor in ANC clinic improved knowledge about IPV and women's rights among pregnant women
Mean (SD) score of learning about women's rights in relationship	Intervention = 2.6 (1.1) Control = 2.0 (1.0) p<0.0001	-				
Women attitudes about VAW (n=1)						
Proportion (n) who justified wife beating, post-intervention	Intervention = 18.3% (49/268) Control = 21.8% (58/267) p=0.33	-	337 women (RCT/1 month)	Haberlan 2016[38]	High	HCP training and ongoing support, 29-minute integrated HIV-IPV consultation, assisted onsite referral had no effect on attitudes about IPV among pregnant women
Women readiness for addressing VAW (n=1)						
Proportion (n) who felt more confident in how deserve to be treated, post-intervention	Intervention = 82% (73/107) Control = 71.6% (73/134) p=0.12	-	337 women (RCT/1 month)	Haberlan 2016[38]	High	HCP training and ongoing support, 29-minute integrated HIV-IPV consultation, assisted onsite referral had no effect on self-confidence among pregnant women
Intermediate effects on health-related behaviour and practices						
VAW enquiry rate (n=2)						
Proportion (n) screened for IPV, post-intervention	Intervention = 76% (81/107) Control = 22% (29/134) p<0.0001	-	337 women (RCT/1 month)	Haberland 2016[38]	High	HCP training and ongoing support, 29-minute integrated HIV-IPV consultation, assisted onsite referral increased IPV enquiry rate
Proportion (n) who discussed IPV	Pre-intervention = 67.3% (201) Post-intervention = 96.5% (387) p<0.01	-	408 HCPs (UBA/6 months)	Jayatilleke 2015[39]	High	4-day HCP training, handbook, external referral increased IPV enquiry rate
Provision of referrals to VAW services (n=2)						
Proportion (n) referred to GBV centre of those disclosed, post-intervention	Intervention = 56% (19/34) Control = 33% (3/9) p=0.28	-	337 women (RCT/1 month)	Haberland 2016[38]	High	HCP training and ongoing support, 29-minute integrated HIV-IPV consultation, referral to IPV counsellor in ANC clinic had no effect on referral rate

Outcomes	Results (95% CI, p)	Effect (95% CI)	N of participants (design/follow up)	Study	Risk of bias	Direction of effect/interpretation
Proportion (n) referred to the medical officer or Heath/IPV services	Pre-intervention = 6.5% (13) Post-intervention = 22.4% (87) p not reported	-	408 HCPs (UBA/6 months)	Jayatilleke 2015[39]	High	4-day training, handbook, external referral had no effect on referral rates to external IPV services
Proportion (n) referred to GBV centre of those disclosed, post-intervention	75% (73/95)	-	1210 women (Cross sectional/7 months)	Undie 2016[57]	High	HCP training, integrated IPV-HIV consultation, assisted referral contributed towards 75% referral rate to on-site GBV centre
Provision of post-rape care (n=2)						
Change in proportion who used a rape kit	Intervention = 59.6% Control = -4.5%	Net effect = 64.1% (46.7; 81.5)	100 HCPs (CBA/12 months)	Abeid 2016[28]	High	5-day training, guidelines, infrastructure improvement contributed towards improvement on 10 out of 18 indicators of post-rape care
Change in proportion who gave prophylactic treatment for STI	Intervention = 10.9% Control = 3.4%	Net effect 7.5% = (-14.5; 29.5)				
Proportion of eligible patients who received emergency contraception	Pre-intervention = 50% Post-intervention = 82% p<0.01	-	60 patients (UBA/3 months)	Smith 2013[54]	High	4-day training, infrastructure improvement contributed towards improvement on 6 out of 10 indicators of post-rape care service
Proportion of eligible patients who received HIV post-exposure prophylaxis	Pre-intervention = 42% Post-intervention = 92% p<0.001	-				
Proportion of eligible patients who received STI prophylaxis and treatment	Pre-intervention = 45% Post-intervention = 96% p<0.01	-				
VAW disclosure rate (n=4)						
Proportion (n) who disclosed IPV of those screened, post-intervention	Intervention = 32% (34/107) Control = 7% (9/134) p<0.0001	-	337 women (RCT/1 month)	Haberland 2016[38]	High	HCP training and ongoing support, 29-minute integrated HIV-IPV consultation, referral to IPV counsellor in ANC clinic increased IPV identification rate
Proportion (n) who identified at least one IPV during past 3 months	Pre-intervention = 73.3% (299) Post-intervention = 98.5% (402) p<0.001	-	408 HCPs (UBA/6 months)	Jayatilleke 2015[39]	High	4-day training, handbook increased IPV identification rate
Proportion who disclosed IPV of those screened	62%	-	102 women (Cross-sectional/3 weeks)	Laisser 2011[45]	High	HCP training, infrastructure improvement, integrated ANC-IPV consultation, external referral contributed towards 62% IPV identification rate
Proportion (n) who disclosed IPV of those screened	8% (95/1210)	-	1210 women (Cross-sectional/7 months)	Undie 2016[57]	High	HCP training, integrated HIV-IPV consultation, assisted onsite referral contributed towards 8% IPV identification rate
VAW referral uptake (n=2)						
Proportion (n) who used GBV centre out of those referred, post-intervention	Intervention = 63% (12/19) Control = 100% (3/3) p=0.52	-	337 women (RCT/1 month)	Haberland 2016[38]	High	HCP training and ongoing support, 29-minute integrated HIV-IPV consultation, referral to IPV

Outcomes	Results (95% CI, p)	Effect (95% CI)	N of participants (design/follow up)	Study	Risk of bias	Direction of effect/interpretation
						counsellor in ANC clinic had no effect on uptake of referrals to on-site GBV centre
Proportion (n) who used GBV centre out of those referred, post-intervention	40% (29/73)	-	1210 women (Cross sectional/7 months)	Undie 2016[57]	High	HCP training, integrated IPV-HIV consultation, assisted onsite referral contributed towards 40% uptake of referrals to on-site GBV centre
Use of SRH services (n=1)						
Proportion (n) who were linked to medical care to receive lab reports on CD4 count and viral load, post-intervention	Intervention = 43.13% (69/160) Control = 38.50% (30/78) p = 0.493	-	166 women (RCT/1 month)	Brown 2018[32]	High	7-minute integrated HIV-IPV consultation over phone had no effect on uptake of HIV services among women with experience of IPV
Safety behaviour (n=2)						
Mean (SD) pre-post difference score of perceived risk and safety	Intervention = 0.33 (3.07) Control = 0.13 (3.05) p=0.278	-	166 women (RCT/1 month)	Brown 2018[32]	High	7-minute integrated HIV-IPV consultation over phone had no effect on perceived risk and safety among HIV-positive women with experience of IPV
Proportion (n) who used safety plan, post-intervention	Intervention = 61.88% (99/160)	-				Most HIV-positive women who received 7-minute integrated IPV-HIV consultation used safety plan and employed at least one safety strategy
Proportion (n) who employed at least one safety strategy	Intervention = 80% (128/160)	-				
Proportion (n) who took an action following the IPV-enhanced HIV counselling, post-intervention	Intervention = 45.5% (25/66) Control = 30.5% (18/79) p=0.073	-	337 women (RCT/1 month)	Haberland 2016[38]	High	HCP training and ongoing support, 29-minute integrated HIV-IPV consultation, referral to IPV counsellor in ANC clinic had no effect on coping behaviour and 7 behaviour indicators of HIV care among pregnant women
Proportion (n) who can ask partner to use a condom, post-intervention	Intervention = 58.3% (35/107) Control = 51.2% (43/134) p=0.31	-				
Health outcomes						
Re-exposure to VAW (n=3)						
Proportion (n) who did not experience IPV upon partner notification of serostatus, post-intervention	Intervention = 96.9% (155/160) Control = 88% (71/79)	OR = 4.37 (1.46; 13.44)	166 HIV-positive women (RCT/1 month)	Brown 2018[32]	High	7-minute integrated HIV-IPV consultation over phone consultation reduced IPV upon partner notification about serostatus among HIV-positive women
Proportion (n) who experienced any IPV, post-intervention	Intervention = 16.0% (43/337) Control = 18.7% (50/351), p=0.43	-	337 pregnant women (RCT/1 month)	Haberland 2016[38]	High	HCP training and ongoing support, 29-minute integrated HIV-IPV consultation, referral to IPV counsellor in ANC clinic had no effect on any IPV since baseline assessment
Mean (SD) danger assessment score	Pre-intervention = 6.02 (2.97) Post-intervention = 2.82 (0.27)	MD = 3.20 (3.56) (2.43; 3.98)	84 women (UBA/3 months)	Matseke 2013	High	HPC training, 30-minute integrated ANC-IPV consultation, external referral contributed towards reduction in potential risk of becoming a victim of femicide among pregnant women

Outcomes	Results (95% CI, p)	Effect (95% CI)	N of participants (design/follow up)	Study	Risk of bias	Direction of effect/interpretation
Harm (n=2)						
Proportion (n) who reported that service had not placed them in greater danger, post-intervention	Intervention = 96.25% (154/160) Control = 93.67% (74/79) p = 0.512	-	166 women (RCT/1 month)	Brown 2018[32]	High	7-minute integrated HIV-IPV consultation over phone did not put HIV-positive women in greater danger
Proportion who reported harmful effects	Intervention = 0	-	337 pregnant women (RCT/1 month)	Haberland 2016[38]	High	HCP training and ongoing support, 29-minute integrated HIV-IPV consultation, referral to IPV counsellor in ANC clinic had no harmful effect on pregnant women

Note. VAW violence against women. GBV gender-based violence. DV domestic violence. IPV intimate partner violence. SRH sexual and reproductive health. STI Sexually transmitted infections. HIV human immunodeficiency virus. ANC antenatal care. HCP health care provider. RCT randomised controlled trial. CBA controlled before-after study. UBA uncontrolled before-after study. CI confidence interval. SD standard deviation. IQR interquartile range. MD mean difference. OR odds ratio.